AI PROTOTYPE · FINAL MASTER PROJECT

Exposed Rebar Inspection with YOLOv8 Segmentation

AI-assisted flagging of visible exposed reinforcement in concrete-inspection images, for review by a qualified structural engineer

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SLIDE 1

Problem and objective

Engineering problem	Loss of concrete cover exposes reinforcement and raises the risk of corrosion and durability loss.
Current practice	Visual review of large inspection photo sets is slow and depends on individual judgement.
Objective	Detect and segment visible exposed rebar in inspection photographs, so images can be flagged for review.
Intended role	Inspection decision support — not autonomous structural diagnosis.

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The AI task: instance segmentation

Input

A photograph of a reinforced-concrete element taken during inspection.

Output

A bounding box, a pixel-level mask and a confidence score for each predicted exposed-rebar region.

Why masks

A mask outlines the predicted steel region more finely than a box. It is a screening output only — not a measure of damaged area, steel loss or repair quantities.

Target class: `exposed_rebar` — reinforcing steel visible because cover is absent or broken. Label visible steel only; exclude surrounding concrete, voids, rust stains and shadows.

Not labelled: cracks, rust staining without visible steel, spalling without visible bars, shadows, mesh, pipes.

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Method and workflow

Model: YOLOv8n-seg instance-segmentation network, trained on an annotated exposed-rebar dataset (Roboflow rebar-exposure-qm020, v4; 1,547 training / 168 validation images).

1	Dataset preparation and annotation check
2	Model training (30 epochs)
3	Validation on 168 held-out images
4	Inference on 8 new inspection photos (one duplicate photo removed)
5	Evidence, weights and code archived in GitHub

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Training configuration

Parameter	Value
Task	Instance segmentation
Model	YOLOv8n-seg (yolov8n-seg.pt)
Epochs	30
Image size	640 × 640 px
Batch size	8
Hardware	NVIDIA Tesla T4 GPU
Random seed	42
Training time	≈ 20 minutes

Reproducibility

- Single Colab notebook reproduces training, validation and inference.
- Fixed seed 42 and recorded hyper-parameters.
- Selected checkpoint best.pt stored in the repository.
- Curves and epoch-level metrics exported as results.png and results.csv.

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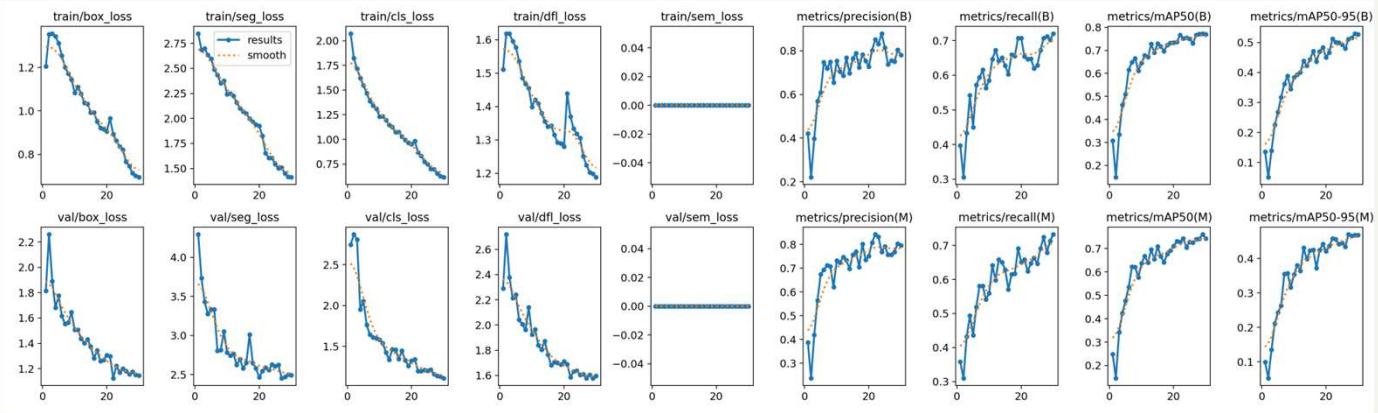
Validation results after 30 epochs

Metric	Boxes	Masks
Precision	0.805	**0.808**
Recall	0.703	**0.717**
mAP@50	0.772	**0.762**
mAP@50–95	0.529	**0.470**

Validation set: 168 images, 229 annotated instances.

Reading the numbers

- Masks are the primary result, because the task needs spatial delineation.
- Recall 0.717: about 28 % of annotated instances are missed — the main risk.
- Lower mask mAP@50–95: outlines are coarse at strict overlap thresholds.



Training and validation curves (results/curves/results.png): losses fall and validation metrics rise overall, with epoch-to-epoch fluctuation. Recall had not clearly plateaued at epoch 30.

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Prediction evidence on new inspection images

Blue boxes and masks show the model's predicted image regions for exposed reinforcement. They support visual screening only: a qualified structural engineer verifies the original image, and no damaged area, repair quantity or structural condition is derived from the masks.

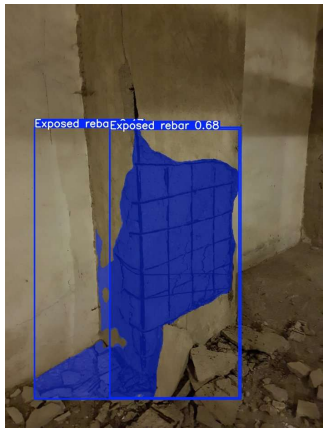


Image 8 — likely true positive

Visible bar grid; mask also covers adjacent spalled concrete and debris.

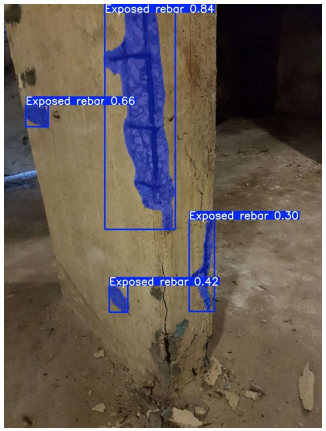


Image 4 — suspected false positives

Spalling, cavities and texture flagged (0.30–0.84); no distinct visible bar.

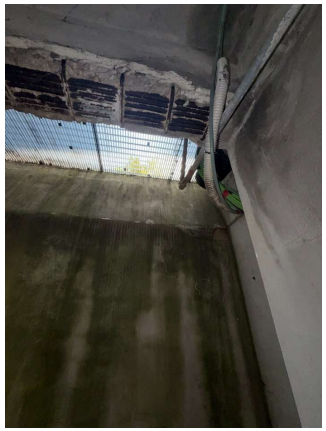


Image 1 — no detection

Bars suspected near the upper edge in low light; no detection ≠ defect-free.

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Engineering value, limitations and next steps

Value

- Faster first-pass review of site photographs
- Consistent visual flagging of suspected exposure
- Prioritises which images an engineer reviews first

Limitations

- Sensitive to lighting, shadows, occlusion and viewpoint
- Confuses cracks, staining and spalling with rebar
- Masks over-extend into damaged concrete; no corrosion severity or repair need

Next steps

- Add crack/spall negatives and close-range soffit images
- Tune confidence threshold and NMS; set minimum image scale
- Validate on a larger independent site set, under engineer review

A detection is an inspection flag, not a structural diagnosis. Every output must be reviewed by a qualified structural engineer.